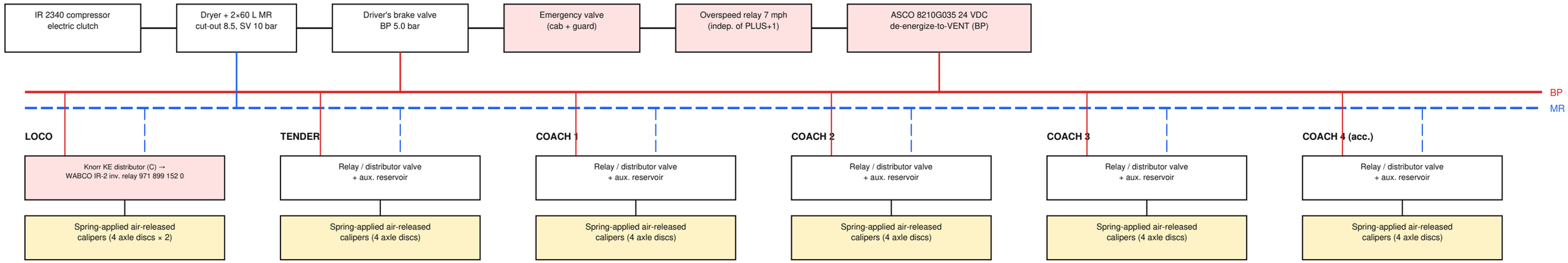
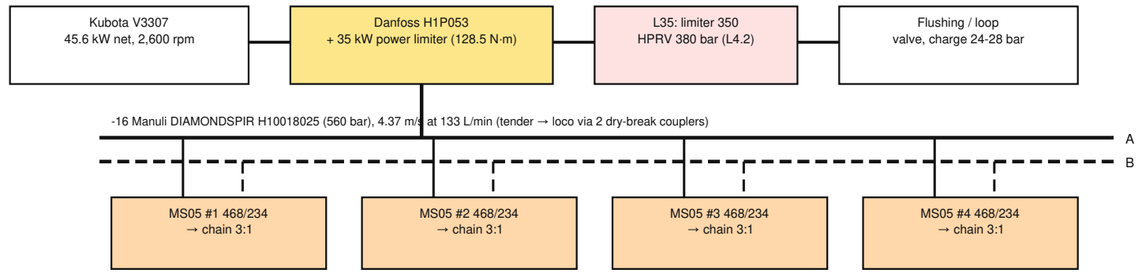
AIR BRAKE (automatic, BP 5.0 bar / 72.5 psi UIC; fail-safe: loss of BP, MR or 24 V applies spring brakes on every axle except the unbraked pony)



BRAKE PERFORMANCE (calc §5)

Case (loaded train, 2% down)	Result	Criterion
Emergency force (tare-based 12% ratio)	67.5 kN	< wet adhesion of braked tare 80.2 kN
Emergency decel. level / 2% down	0.828 / 0.632 m/s²	-
Emergency stop from 6 mph (governed)	9.7 m	≤ 20 m (ASSUMED)
Emergency stop from 7 mph (overspeed trip)	12.4 m	≤ 30 m
Service stop (70%) from 6 mph	13.4 m	≤ 30 m
Parking: all spring brakes, FS 1.5 (T04 = parking/holding only, 0 in motion)	24.0 kN req. vs 64.1	OK - loco alone NG → rule + chocks
Loco axle-disc clamp per caliper (2/disc)	19.8 kN	spec ≥ 22 kN
In-train buff, loco-tender	4.1 kN	≤ 70 kN

HYDRAULIC TRANSMISSION (closed loop, ISO 4413)



FAIL-SAFE LOGIC (loco, rev L4 - PE to confirm)

BP drop (driver valve, emergency valve, train parting) → KE raises C → IR-2 EXHAUSTS the spring chambers → loco springs apply.
Loss of 24 V, or Pilz PNOZ s30 7 mph trip → ASCO 8210G035 de-energizes → BP VENTS → emergency on every vehicle; the same s30 safe output drops the H1P053 CCO (control cut-off) → pump to neutral. Loss of MR / IR-2 supply → chambers exhaust → APPLY.
Release needs BP ≥ 4.8 bar AND MR AND 24 V healthy. IR-2 supply regulated ≤ 6 bar (Coremo release max) - verify IR-2 data sheet.

DRIVE NOTES

Traction (L4): 2% start need 32.61 kN; 37.67 kN at 350 bar; NO flow divider - PLUS+1 traction control (pony ref., destroke); NO SAND (deleted). Reverse pull cap 28.9 kN; reverse full-train starts ≤ 1.20 % (L4.2). Sanders DELETED.
GOVERNED 6 mph: H1P053 stroke stop 42 cc sealed + HARD tamper-sealed 2,600 rpm engine limit (hardware 7.64 mph). 7 mph trip: 2-channel speed relay independent of PLUS+1 → BP vent + pump neutral.
ANY TRIP / 24 V loss (L4.2): swash ramp ≥ 3 s + loop BYPASS normally open = FAIL-TO-FLOAT (1st-gear float sized); 2-ch T04 switch vents BP first;
T04 choke Ø2.4 + 1 L acc. + Ø1.2 orifice per port, hard/4SH lines downstream (run-in ≤ 33.2 kN). Direction change only at zero speed.
T04 RELEASE (L4.2): H1 CHARGE port M3 → check + 10 µm → CDK 20 bar reducing/relieving → NV 3/2 → choke → 4 × T04 (loop shuttle DELETED). Release ≤ 8 s (met).
Manual release: check → 6.0 L bladder acc. (13.5 bar) + 12 cm³ hand pump → guarded padlocked 3/2 (limit sw. → traction inhibit); T04 M12 screws last resort.
35 kW anti-stall (34.5 kW while the compressor clutch is in; clutch only at low demand or MR < 7.0 bar). H1P053 with CCO (to the s30 trip); external charge filter HYDAC DF 60.
TOWING: tow valve (KHB3K DN06, 500 bar) open, T04 release by 6 L acc. + hand pump, max 1 mph (E11-S5 ruling 6). DESCENTS in FLOAT (bypass open, ≥ 3 s ramp) on friction brakes sized for 2 %. Draft gear: preloaded, k 20-100 kN/mm, renew at 1.0 mm TOTAL play. Loco brake ≤ 0.69 m/s².
LOCO RELEASE ≤ 4 s (L4.2: 2.22 / 2.36 s): 8 × 1.0 L calipers; C_spec ≥ 3.2 per branch; QEV option B (maker-cert. ≥ 15 bar): relieving regulator 1/2 in C ≥ 10 @ 6.0 bar → 1/2 in relay C ≥ 8 (piloted by IR-2) →
3/4 in trunk ID ≥ 19 C ≥ 8 → 4 × 1/2 in hose ≤ 2 m C ≥ 2.0 to both calipers; quick-exhaust per axle; release-proof 5.1 bar + coach release 2-ch (E3.6h) → traction enable. Test ≤ 4.0 s at MR cut-in.
Loop hose Manuli DIAMONDSPIR H10018025 (560 bar WP, ISO 18752-C, OD 41.2, min bend 350 ≤ tender R70 394.3). Loop parts ≥ 420 bar WP ≥ HPRV 380;
spike ~430 covered by ISO 18752 / ISO 6605 impulse rating; strength checked at 430; whip restraints; system test per ISO 4413 (PE), not above 420.

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Brake, air and hydraulic schematic

Sheet B-601 · REV L4.2 · 2026-10-28 · NTS

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

REV 2026-11-02 cleanup: figures per data/mass_ledger.json and status.html (closed rulings). Speed: 6 mph governed limit · 6.5 mph HMI warning · 7 mph independent trip · 3 mph max if cameras fail. Not PE-reviewed.